# Journal Pre-proof

Incidence, Management, and Outcomes following Elbow Ulnar Collateral Ligament Injuries in Non-Quarterback Professional Football Players

Neel K. Patel, MD, Adeeb Hanna, DO, William L. Johns, MD, Arsh S. Dhanota, MD, Christopher C. Dodson, MD, Matthew Pepe, MD

PII: S2666-6383(25)00131-8

DOI: https://doi.org/10.1016/j.jseint.2025.04.010

Reference: JSEINT 1265

To appear in: *JSES International*

Received Date: 6 April 2025

Accepted Date: 11 April 2025

Please cite this article as: Patel NK, Hanna A, Johns WL, Dhanota AS, Dodson CC, Pepe M, Incidence, Management, and Outcomes following Elbow Ulnar Collateral Ligament Injuries in Non-Quarterback Professional Football Players, *JSES International* (2025), doi: https://doi.org/10.1016/j.jseint.2025.04.010.

This is a PDF file of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability, but it is not yet the definitive version of record. This version will undergo additional copyediting, typesetting and review before it is published in its final form, but we are providing this version to give early visibility of the article. Please note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2025 Published by Elsevier Inc. on behalf of American Shoulder and Elbow Surgeons.

Journal Pre-proof

# Incidence, Management, and Outcomes following Elbow Ulnar Collateral Ligament Injuries in Non-Quarterback Professional Football Players

**Authors:** Neel K. Patel MD[1], Adeeb Hanna DO[1], William L. Johns MD[1], Arsh S. Dhanota MD[2], Christopher C. Dodson MD[1], Matthew Pepe MD[1]

*Institution*

[1] Rothman Orthopaedic Institute at Thomas Jefferson University Hospitals, Philadelphia, PA, USA

[2] Department of Sports Medicine, Perelman School of Medicine at the University of Pennsylvania, Philadelphia, PA, USA

**Corresponding Author:** Matthew Pepe, MD Matthew.pepe@rothmanortho.com

Rothman Orthopaedic Institute at Thomas Jefferson University Hospitals, Philadelphia, PA, USA

2500 English Creek Ave, Building 1300 Egg Harbor Township, NJ 08234, USA

**Disclaimers:**

**Funding:** No funding was disclosed by the authors.

**Conflicts of interest:** The authors, their immediate families, and any research foundation with which they are affiliated have not received any financial payments or other benefits from any commercial entity related to the subject of this article.

Thomas Jefferson University Hospitals Institutional Review Board approved this study. IRB Approval Number: iRISID- 2022-1115

Journal Pre-proof

Journal Pre-proof


# Incidence, Management, and Outcomes following Elbow Ulnar Collateral Ligament Injuries in Non-Quarterback Professional Football Players


## ABSTRACT

**Background:** Ulnar collateral ligament (UCL) injuries can cause significant disability, especially in overhead throwing athletes. A majority of studies in the literature focus on the effects of UCL injuries in baseball players, but little is known about the severity and outcomes of these injuries in elite non-throwing contact athletes. The purpose of this study was to characterize the severity of UCL injuries and describe management outcomes in non-throwing professional football players.

**Methods:** A retrospective review was conducted using the National Football League (NFL) Injury Surveillance System (ISS) on non-quarterback players that had a UCL injury from 2015-2022. Player demographics, player position, severity and mechanism of injury, type of management, and time to return to play (RTP) were collected. The severity of injury was classified based on laxity on clinical examination and graded on a scale of I-III. Data was analyzed to determine the number of injuries that occurred per year, per position, and per setting to determine differences in time from play that was missed based on injury severity. Statistical significance was set a $p < 0.05$.

**Results:** A total of 268 non-quarterback NFL players sustained isolated UCL injuries between 2015-2022, including 33.5±4.2 injuries per year. Offensive and defensive linemen were the most common positions to sustain isolated UCL injuries, accounting for 56.0% (150/268 total injuries total) of all non-quarterback UCL injuries in the NFL. Ninety-eight percent of player with isolated UCL injuries were treated non-operatively with only 4 players requiring surgical

Journal Pre-proof


treatment of their injury. Seventy-three percent of injuries were Grade 1 sprains, 16.8% were Grade II sprains, and 10.1% were Grade III sprains. Seventy-two percent of athletes did not miss any time due to their isolated UCL injury. In the 28% of players that did miss time, the mean time missed was 5.4±24.2 days. There was a significant difference in the time missed between Grade I and Grade II injuries (1.1 days vs. 12.4 days; $p < 0.05$) and Grade I and Grade III injuries (1.1 days vs. 26.3 days; $p < 0.05$).

**Conclusions:** Isolated UCL injuries are relatively uncommon injuries in non-quarterback NFL players. The vast majority of injuries are Grade I sprains and are treated non-operatively. Grade II and Grade III injuries result in significantly increased time missed compared to Grade I injuries. This information can be useful when counseling players regarding prognosis and timing of RTP based on severity of injury.

**Level of Evidence/Study Design:** Level IV; Descriptive Epidemiology Study

**Keywords:** Ulnar collateral ligament; Professional Football; Contact athletes; Elbow; National Football League; Return to Play

Ulnar collateral ligament (UCL) injuries are common elbow injuries in high level athletes across many sports such as baseball, wrestling, and gymnastics[14, 24]. The literature to date has primarily focused on UCL injuries in throwing athletes, particularly in baseball pitchers[10, 23]. Throwing athletes generally achieve excellent outcomes after UCL reconstruction and repair with return to play (RTP) at the same or higher level between 83-95%[2, 20, 23]. In contrast, there is limited information in the literature regarding outcomes of UCL injury and management in non-throwing, contact athletes[18].

Journal Pre-proof

Journal Pre-proof


Several studies have evaluated the effects and outcomes of UCL injury in professional baseball players. One study characterized injuries in Major League Baseball (MLB) pitchers and found that pitchers had a significantly higher rate of UCL injury in the first 3 months of the season and the risk of injury correlation with pitching velocity[6]. The prevalence of UCL reconstruction in professional baseball pitchers is 16% compared to only 3% in position players, with the prevalence in both groups increasing with time[5, 13]. Additionally, 83% of professional pitchers who underwent UCL reconstruction were able to return to pitching in the MLB[8]. For partial injuries in professional baseball players, the overall return to play rate was 85%[22]. Despite the extensive number of studies on UCL reconstruction in professional baseball players, there is limited information regarding lower grade injuries in the in elite non-throwing athletes.

Non-throwing contact athletes appear to have different injury mechanisms and tear patterns due to different strains imparted upon the elbow during sport participation compared to throwing athletes. A majority of UCL injuries in overhead throwing athletes are a result of chronic attenuation due to repetitive valgus stress during the throwing motion, while UCL injuries in contact athletes are usually a result of acute tears or avulsions in an otherwise healthy ligament[21]. Frequently described injury mechanisms in non-throwing athletes are generally acute events including valgus forces applied through the elbow during blocking in National Football League (NFL) linemen, arm bars in martial arts, and back handsprings in gymnastics[11, 12]. In their evaluation of NFL quarterbacks, Dodson et al posited that contact athletes, even non-throwers, have a predisposition towards acute, high-contact loads about the elbow and are thus subject to a different spectrum of UCL injury patterns irrespective of throwing[7]. Despite this predisposition and the potential differences in optimal treatment strategies compared to predominantly throwing

Journal Pre-proof


athletes, there are currently no studies that characterize the incidence and management of isolated UCL injuries in non-quarterback NFL players.

Thus, the purpose of this study was to determine incidence, management, and timing of RTP following isolated UCL injuries in NFL players. It was our hypothesis that UCL injuries would be more frequent among lineman compared to skill position players and a majority of players would be treated non-operatively.

## MATERIALS AND METHODS

Approval for this study was obtained from the Thomas Jefferson University Hospital Institutional Review Board (IRB) (iRISID- 2022-1115) and through the NFL-NFL Players Association (NFLPA) Player Scientific and Medical Research Approval Process[15]. A retrospective review was conducted using the NFLISS on non-quarterback NFL players who sustained a UCL injury from 2015 to 2022. The injury in the NFLISS data is based on the primary diagnosis/diagnoses made by the medical and training staff of each NFL team during the season. This data is submitted to a central NFL electronic medical record. The NFLISS has been used in many previous studies to report injury rates in the NFL. The NFLISS documented the presence and grade of UCL injuries based upon physical examination and radiographic/MRI findings as documented in the players' records in the league-wide electronic medical record. Grading of the severity of the injury was based upon physical examination or MRI findings. Players with any additional elbow injury (lateral collateral ligament injury, triceps tendon rupture, etc.) were excluded from the study.

Once the final cohort of patients was identified, information was collected on player demographics, player position at the time of injury, mechanism of injury (contact or non-

Journal Pre-proof


contact), timing of injury (pre-season, regular season, playoffs), setting of injury (practice vs. game), severity of injury, treatment method (non-operative vs. operative), and time missed (days missed were supplied by the NFLPA, not games). The severity of injury was classified based on laxity on clinical examination and graded on a scale of I-III[3, 9, 21]. Grade I represented a sprain, Grade II represented a severe sprain or partial tear, and Grade III represented a complete tear of the ligament. The time missed was based on the time between the injury and full, unrestricted participation in practices and games.

The data was analyzed to determine the number of injuries that occurred per year, per setting, and per position. Additionally, the time missed was analyzed as a whole and based on severity of injuries and t-tests or Mann-Whitney-U tests were used to determine significant differences in time missed based on the severity of injury. Statistical significance was set at $p < 0.05$.

## RESULTS

### *Cumulative Characteristics of UCL Injuries*

After screening the NFLISS, a total of 268 non-quarterback NFL players had isolated UCL injuries between the years 2015-2022. The number of injuries per year was relatively stable over that course of time with 33 injuries in 2015, 24 injuries in 2016, 35 injuries in 2017, 38 injuries in 2018, 33 injuries in 2019, 36 injuries in 2020, 33 injuries in 2021, and 35 injuries in 2022 with a mean of 33.5±4.2 injuries per year. Offensive and defensive linemen (60/268) were the most common non-quarterback positions to sustain an isolated UCL injury, accounting for 56.0% (150/268 total injuries) of the overall injuries with offensive linemen being the single most common position at 33.5% of the injuries (90/268). The other positions injured were linebackers (39 injuries), defensive secondary (26 injuries), running backs (11 injuries), tight ends (11

Journal Pre-proof


injuries), wide receivers (4 injuries), and special teams players (22 injuries). Position was not documented for 5 of the injuries. When assessing each grade of UCL injury by position, offensive linemen had the highest percentage of injuries in each severity grade accounting for 32.9% of Grade I injuries, 35.5% of Grade II injuries, and 33.3% of Grade III injuries. The linebacker position was the only position that had an increase in the percentage of injuries with increasing injury severity accounting for only 12.7% of Grade I injuries and 15.6% of Grade II injuries, but more than a quarter (25.9%) of Grade III injuries.

A majority of UCL injuries (73.5%) occurred during a game (197 injuries), while 24.6% of UCL injuries occurred during practice (66 injuries) and 5 UCL injuries did not have a documented setting of injury. With regard to timing of these injuries, 72.8% of the injuries occurred in the regular season, 23.9% occurred in the pre-season, and 3.4% occurred in the post-season. Of the injuries that had a documented mechanism of injury, all injuries occurred due to contact, either directly with another player or due to indirect contact (42.7%, 95/222 injuries). Forty-six injuries did not have a documented mechanism of injury.

*Treatment and Time Missed for UCL Injuries*

A vast majority of players (98.5%) with isolated UCL injuries were treated non-operatively. The exact mode or duration of non-operative treatment was not specified in the NFLISS. A total of 4 players (3 linebackers and 1 special teams player) required surgical treatment of their isolated UCL injuries (Table 1). Details regarding the time, attempted nonoperative treatment modalities, surgical indications, and specific procedures that they underwent were not available.

Overall, 72% of players did not miss any time due to their isolated UCL injury. In the 28% of players that did miss time, the average time missed was 5.4±24.2 days. When characterizing the

Journal Pre-proof

Journal Pre-proof


injuries by severity, it was found that 73% of injuries were Grade 1 sprains (196 injuries), 16.8% were Grade II sprains (45 injuries), and 10.1% were Grade III sprains (27 injuries). The breakdown of time missed based on injury severity is shown in Table 2. There was a significant difference in the time missed between Grade I and Grade II injuries (1.1 days vs. 12.4 days, respectively; $p < 0.05$) and Grade I and Grade III injuries (1.1 days vs. 26.3 days, respectively; $p < 0.05$). There was no statistically significant difference in time missed between Grade II and Grade II injuries (12.4 vs 26.3 days; $p > 0.05$).

## DISCUSSION

While UCL injuries have been known to be common in baseball players, especially pitchers, little is known regarding the characteristics of UCL injuries in non-quarterback professional football players. The findings of this study show that approximately 34 isolated UCL injuries in non-quarterback professional football players occur each season with a large majority of injuries being Grade I injuries. Offensive and defensive lineman are the positions in which players most commonly sustained a UCL injury, and linebackers most commonly sustained higher grade 3 injuries. Most players were able to be successfully RTP with non-operative treatment (98.5%) and 72% of players did not miss any time due to their injuries. However, Grade II and III injuries did have a significantly higher time missed compared to Grade I injuries.

An analysis of UCL injuries in the National Collegiate Athletic Association (NCAA) across 25 varsity sports in a 5-year span found that a majority of UCL injuries occurred as a result of contact (76.1%) and football accounted for 32.1% of overall UCL injuries in NCAA over the course of a single season[14]. Of those UCL injuries that occurred in football players, all injuries were due to a contact mechanism as was seen in the population of this study of NFL players[14]. In

Journal Pre-proof


the NCAA football players, 90.6% of the players with reported time to RTP data returned within 21 days of the injury (mean time to return of approximately 10 days), which is consistent with the RTP seen in the NFL population (93.7%; mean time to return of approximately 6 days)[14]. Additionally, no football players in the NCAA population underwent surgical treatment for their injuries[14]. Overall, the injury mechanism, treatment, and RTP timeline is similar in the NCAA football and NFL player populations.

Recently there has been an increased interest in outcomes of UCL reconstruction and repair in non-throwing and contact athletes[19]. Rothermich et al investigated outcomes of UCL reconstructions and repairs in a mix of youth sport, high-school, college, and professional-level non-throwing athletes and found a nearly 95% RTP rate, which is comparable to the rate of RTP in throwing athletes[10, 19]. In the cohort of NFL players in this study, all were able to RTP after non-surgical and surgical management. The average time of RTP for the players in this study after surgical treatment was 4.7 months, which is a much shorter period of time compared to that reported by Rothermich et al at 7.4 months in the repairs and 10.0 months in the reconstructions[19]. While not specified by the NFLISS, given that 2 players returned within 2 months, we speculate that they underwent a UCL repair and the other 2 players who missed more than 6 months likely underwent reconstruction. However, the differences in time to RTP between our study and that reported by Rothermich et al are difficult to interpret since the Rothermich et al study is a mixed cohort of athletes, not isolated to professional football players. Furthermore, there may also be a difference is definition of RTP due to the fact that the NFLISS measures a RTP as full participation in practice or games, which is likely more objective than patient-reported return to competition as in the previously mentioned study.

Journal Pre-proof

Journal Pre-proof


When comparing the treatment of contact injuries to those of overhead throwing, the primary treatment in throwing athletes is UCL reconstruction, as evidenced by the significant prevalence of the surgery in professional pitchers[1, 7]. Given the difference in mechanism, a majority of contact injuries can be treated non-operatively as seen in this cohort with 98.5% being treated non-operatively. The severity of the injuries did predict increased time to RTP, though even with complete Grade III injuries a majority of players were able to be treated non-operatively. Furthermore, even the players that underwent operative management were able to RTP at a much earlier time point than is typically seen in a throwing athlete after UCL reconstruction or repair.

The mode and duration of various nonoperative treatment modalities was not specified by the NFLISS. However, in the general population initial nonoperative treatment often consists of a combination of rest, ice, anti-inflammatories, and other pain modalities along with bracing or immobilization of the elbow and potentially the use of biologic treatments such as platelet-rich plasma (PRP). Further studies are needed to evaluate the effects of different non-operative treatment modalities on outcomes and time to RTP. For example, the use of PRP for non-operative treatment of UCL injuries in throwing athletes has shown mixed results with no improvement in RTP outcomes, but its role in non-operative treatment in contact athletes is still not known[4, 17]. One case series suggested that the use of PRP may be beneficial in National Hockey League players with high-grade UCL injuries[16]. Of note, the average RTP in those 3 NHL players was 36 days, which is similar to the 26 days seen with Grade III injuries in NFL players[16].

This study has limitations including the retrospective nature of the review based on the NFLISS. First, while this study is able to report on time to return to sport participation, the nature of the study does not allow for an evaluation of return to *performance* and an understanding of the

Journal Pre-proof

Journal Pre-proof


impact of a UCL injury on the performance of a non-throwing NFL athlete remains to be elucidated. Additionally, the concept of time to RTP could be heavily influenced by external factors such as timing of injury in the season – thus one athlete with a grade 2 sprain treated nonoperatively may have missed 100 days of sport participation as a result of an injury later in his respective season requiring a return the following offseason and thus a longer reported time to RTP. Third, the present study does not provide any information regarding the utility of bracing, PRP, or other nonoperative treatment options that are utilized in the overwhelming majority (98.5%) of these contact athletes. Magnetic resonance imaging results for the injuries were not available, and thus radiographic details such as tear type, tear location, extent of tears, or other concomitant injuries as shown on MRI are not available. Furthermore, in the event of surgical intervention there is no detailed information regarding surgical indications, treating surgeon, or specific techniques (graft harvest site, repair vs. reconstruction, reconstruction technique, etc.) that were performed. Additionally, patient reported outcomes with longer term follow-up is not available through the NFLISS. Lastly, the NFLISS relies upon each medical staff to report injuries as they occur. Thus, it is possible that the data presented in this study may under-represent the true incidence of UCL injuries in the NFL given the nature of such reporting systems.

## CONCLUSIONS

Isolated UCL injuries are relatively uncommon injuries in non-quarterback NFL players. The vast majority of injuries are Grade I sprains and are treated non-operatively. Grade II and Grade III injuries result in significantly increased time missed compared to Grade I injuries. This information can be useful when counseling players regarding prognosis and timing of RTP based on severity of injury.

Journal Pre-proof

Journal Pre-proof


**REFERENCES:**

1 Azar FM, Andrews JR, Wilk KE, Groh D. Operative treatment of ulnar collateral ligament injuries of the elbow in athletes. Am J Sports Med 2000;28:16-23.

2 Cain EL, Jr., Andrews JR, Dugas JR, Wilk KE, McMichael CS, Walter JC, 2nd et al. Outcome of ulnar collateral ligament reconstruction of the elbow in 1281 athletes: Results in 743 athletes with minimum 2-year follow-up. Am J Sports Med 2010;38:2426-2434. 10.1177/0363546510378100

3 Carr JB, Camp CL, Dines JS. Elbow Ulnar Collateral Ligament Injuries: Indications, Management, and Outcomes. Arthroscopy: The Journal of Arthroscopic & Related Surgery 2020;36:1221-1222. https://doi.org/10.1016/j.arthro.2020.02.022

4 Chauhan A, McQueen P, Chalmers PN, Ciccotti MG, Camp CL, D'Angelo J et al. Nonoperative Treatment of Elbow Ulnar Collateral Ligament Injuries With and Without Platelet-Rich Plasma in Professional Baseball Players: A Comparative and Matched Cohort Analysis. Am J Sports Med 2019;47:3107-3119. 10.1177/0363546519876305

5 Conte SA, Fleisig GS, Dines JS, Wilk KE, Aune KT, Patterson-Flynn N et al. Prevalence of Ulnar Collateral Ligament Surgery in Professional Baseball Players. Am J Sports Med 2015;43:1764-1769. 10.1177/0363546515580792

6 DeFroda SF, Kriz PK, Hall AM, Zurakowski D, Fadale PD. Risk Stratification for Ulnar Collateral Ligament Injury in Major League Baseball Players: A Retrospective Study From 2007 to 2014. Orthop J Sports Med 2016;4:2325967115627126. 10.1177/2325967115627126

Journal Pre-proof

Journal Pre-proof


7 Dodson CC, Thomas A, Dines JS, Nho SJ, Williams RJ, 3rd, Altchek DW. Medial ulnar collateral ligament reconstruction of the elbow in throwing athletes. Am J Sports Med 2006;34:1926-1932. 10.1177/0363546506290988

8 Erickson BJ, Gupta AK, Harris JD, Bush-Joseph C, Bach BR, Abrams GD et al. Rate of return to pitching and performance after Tommy John surgery in Major League Baseball pitchers. Am J Sports Med 2014;42:536-543. 10.1177/0363546513510890

9 Ford GM, Genuario J, Kinkartz J, Githens T, Noonan T. Return-to-Play Outcomes in Professional Baseball Players After Medial Ulnar Collateral Ligament Injuries:Comparison of Operative Versus Nonoperative Treatment Based on Magnetic Resonance Imaging Findings. The American Journal of Sports Medicine 2016;44:723-728. 10.1177/0363546515621756

10 Jack RA, 2nd, Rao S, D'Amore T, Willier DP, 3rd, Gallivan R, Cohen SB et al. Long-Term Sports Participation and Satisfaction After UCL Reconstruction in Amateur Baseball Players. Orthop J Sports Med 2021;9:23259671211027551. 10.1177/23259671211027551

11 Kenter K, Behr CT, Warren RF, O'Brien SJ, Barnes R. Acute elbow injuries in the National Football League. J Shoulder Elbow Surg 2000;9:1-5. 10.1016/S1058-2746(00)80023-3

12 Koh TJ, Grabiner MD, Weiker GG. Technique and ground reaction forces in the back handspring. The American Journal of Sports Medicine 1992;20:61-66.

Journal Pre-proof

Journal Pre-proof


13 Leland DP, Conte S, Flynn N, Conte N, Crenshaw K, Wilk KE et al. Prevalence of Medial Ulnar Collateral Ligament Surgery in 6135 Current Professional Baseball Players: A 2018 Update. Orthop J Sports Med 2019;7:2325967119871442. 10.1177/2325967119871442

14 Li NY, Goodman AD, Lemme NJ, Owens BD. Epidemiology of Elbow Ulnar Collateral Ligament Injuries in Throwing Versus Contact Athletes of the National Collegiate Athletic Association: Analysis of the 2009-2010 to 2013-2014 Seasons. Orthop J Sports Med 2019;7:2325967119836428. 10.1177/2325967119836428

15 Mack C, Matava M, Zeidler K, Sills A, Solomon G. Clinical Research in the National Football League: The Player Scientific and Medical Research Protocol. Curr Sports Med Rep 2020;19:168-174. 10.1249/jsr.0000000000000711

16 McCrum CL, Costello J, Onishi K, Stewart C, Vyas D. Return to Play After PRP and Rehabilitation of 3 Elite Ice Hockey Players With Ulnar Collateral Ligament Injuries of the Elbow. Orthop J Sports Med 2018;6:2325967118790760. 10.1177/2325967118790760

17 Mills FBt, Misra AK, Goyeneche N, Hackel JG, Andrews JR, Joyner PW. Return to Play After Platelet-Rich Plasma Injection for Elbow UCL Injury: Outcomes Based on Injury Severity. Orthop J Sports Med 2021;9:2325967121991135. 10.1177/2325967121991135

18 Ramos N, Limpisvasti O. UCL Injury in the Non-throwing Athlete. Curr Rev Musculoskelet Med 2019;12:527-533. 10.1007/s12178-019-09590-2

Journal Pre-proof

Journal Pre-proof


19 Rothermich MA, Pharr ZK, Mundy AC, Fleisig GS, Mussell EA, Ryan MK et al. Clinical Outcomes of Ulnar Collateral Ligament Surgery in Nonthrowing Athletes. Am J Sports Med 2022;50:3368-3373. 10.1177/03635465221120654

20 Thomas SJ, Paul RW, Rosen AB, Wilkins SJ, Scheidt J, Kelly JDt et al. Return-to-Play and Competitive Outcomes After Ulnar Collateral Ligament Reconstruction Among Baseball Players: A Systematic Review. Orthop J Sports Med 2020;8:2325967120966310. 10.1177/2325967120966310

21 Vaswani R, White A, Dines J. Medial Ulnar Collateral Ligament Injuries in Contact Athletes. Curr Rev Musculoskelet Med 2022. 10.1007/s12178-022-09785-0

22 Walker CM, Genuario JW, Houck DA, Murayama S, Mendez H, Noonan TJ. Return-to-Play Outcomes in Professional Baseball Players After Nonoperative Treatment of Incomplete Medial Ulnar Collateral Ligament Injuries: A Long-Term Follow-up Study. Am J Sports Med 2021;49:1137-1144. 10.1177/0363546521996706

23 Yoshida M, Takenaga T, Takeuchi S, Takaba K, Tsuchiya A, Fukuyoshi M et al. Ulnar Collateral Ligament Reconstruction With the Modified Jobe Technique in Baseball Pitchers. Orthop J Sports Med 2021;9:2325967121998339. 10.1177/2325967121998339

24 Zaremski JL, McClelland J, Vincent HK, Horodyski M. Trends in Sports-Related Elbow Ulnar Collateral Ligament Injuries. Orthop J Sports Med 2017;5:2325967117731296. 10.1177/2325967117731296

**Table Legend**

Journal Pre-proof

Journal Pre-proof


**Table 1:** Injury to the ulnar collateral ligament of the elbow in professional non-quarterback football players reported in the NFL Injury Surveillance System from 2015 to 2022 that required surgical treatment.

**Table 2:** Missed time due to injuries to the ulnar collateral ligament (UCL) based on injury severity and injury setting

Journal Pre-proof

Journal Pre-proof

| Player | Diagnosis | Player Position | Setting of Injury | Mechanism | Days Missed |
|---|---|---|---|---|---|
| 1 | UCL Sprain (Grade III) | Linebacker | Regular Season Game | Contact | 51 |
| 2 | UCL Sprain (Grade II) | Linebacker | Regular Season Game | Contact | 42 |
| 3 | UCL Sprain (Grade III) | Linebacker | Regular Season Game | Contact | 184 |
| 4 | UCL Sprain (Grade III) | Special Teams | Regular Season Game | Contact | 296 |

**Table 1.** Injury to the ulnar collateral ligament of the elbow in professional non-quarterback football players reported in the NFL Injury Surveillance System from 2015 to 2022 that required surgical treatment.

Journal Pre-proof

| Severity of Injury | UCL Injuries Resulting in Missed Time | Average Time Missed (days) | Regular Season Injury |
|---|---|---|---|
| Grade I | 13.3% | 1.1 | 71.2% (141/196) |
| Grade II | 35.6% | 12.4 | 73.3% (33/45) |
| Grade III | 63.0% | 26.3 | 77.7% (21/27) |

**Table 2.** Missed time due to injuries to the ulnar collateral ligament (UCL) based on injury severity and injury setting